

PRACTICAL WEEK-2

Session-2:

Task -1 :

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

CODE:

```
D: > OSSP_Programs > Practical > Week2 > C task1.c
1  #include <stdio.h>
2  #include <fcntl.h>
3  #include <unistd.h>
4  #include <stdlib.h>
5
6  int main() {
7      int source, destination;
8      char buffer[1024];
9      ssize_t bytesRead;
10
11     source = open("File1.txt", O_RDONLY);
12
13     if (source < 0) {
14         perror("Error opening source file");
15         return 1;
16     }
17
18     destination = open("File2.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
19
20     if (destination < 0) {
21         perror("Error opening destination file");
22         close(source);
23         return 1;
24     }
25
26     while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
27         write(destination, buffer, bytesRead);
28     }
29
30     close(source);
31     close(destination);
32
33     printf("File copied successfully!\n");
34
35     return 0;
36 }
37
```

OUTPUT:

```
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week1$  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week1$ cd /mnt/d/OSSP_2520080069/Practical/Week2  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ ls  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ nano task1.c  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ echo "This is the content of File1." >  
File1.txt  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ cat File1.txt  
This is the content of File1.  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ gcc task1.c -o task1  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ ./task1  
File copied successfully!  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ cat File2.txt  
This is the content of File1.  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ █
```

OBSERVATION:

The program successfully copied the contents of File1.txt to File2.txt using the open(), read(), write(), and close() system calls. The system calls provide an interface between the user program and the kernel, allowing the program to access and modify files managed by the operating system. The successful execution demonstrated basic file handling and the transition between user space and kernel space.

Task -2 :

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

OUTPUT:

```

close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libcprz2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\3\6-\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 832") = 832
statfs({st_mode=S_IFREG|0644, st_size=99956, ...}) = 0
mmap(NULL, 78112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0x76d50f6c2000
mmap(0x76d50f6c5000, 507904, PROT_READ|PROT_EXEC, MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x76d50f6c5000
mmap(0x76d50f741000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xf7000) = 0x76d50f741000
mmap(0x76d50f76c000, 8192, PROT_WRITE|MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x76d50f76c000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x76d50f6c0000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x76d50f6bd000
arch_prctl(MODR_CTL, 0x76d50f6bb00) = 0
set_tid_address(&0x76d50f6be12) = 263
set_robust_list(&0x76d50f6bdd0, 24) = 0
rseq(&0x76d50f6bdba0, 0x20, 0, 0x53053053) = 0
mprotect(0x76d50f6be000, 16384, PROT_READ) = 0
mprotect(0x76d50f76c000, 4096, PROT_READ) = 0
mprotect(0x76d50f892000, 4096, PROT_READ) = 0
mprotect(0x76d50f8c0000, 4096, PROT_READ) = 0
mprotect(0x76d50f8f1000, 4096, PROT_READ) = 0
mprotect(0x6080c2943000, 146272, PROT_READ) = 0
mprotect(0x76d50f93f000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_SYSCALL, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\x8b\xcc\x64\xee\x71\x36\x9b\xid*", 8, GRND_NONBLOCK) = 8
munmap(0x76d50f8f5000, 20807) = 0
fcntl("/sys/fs/cgroup", F_GETFSYSFS_MAGIC, f_bsize=4096, f_bfree=0, f_bavail=0, f_ffree=0, f_fsid={val=[0x64532b78, 0xbca775bf]}, f_nameLen=255, f_fsize=4096, f
flags={SA_NOINHERIT|NOINHERIT|NOINHERIT|NOINHERIT} = 0
statfs("/selinux", 0x7fffc2b71a0) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x6080e7c20000
brk(0x6080e7c41000) = 0x6080e7c41000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tcsysfs\nnodev\ttmpfs\nnodev\tbd.... 1024") = 342
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2.... 1024") = 1024
read(3, "e 0 nodevs /dev/pts devpts rw.... 1024") = 1024
read(3, ", access=cclient,msize=65536,trans.... 1024") = 1024
read(3, "des=1024,nodev=noswap 0 /ntmp.... 1024") = 262
read(3, "... 1024") = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
fcntl(3, F_GETFD) = 0
fcntl(4, F_GETFD) = 0
fcntl(5, F_GETFD) = 0
fcntl(6, F_GETFD) = 0
fcntl(7, F_GETFD) = 0
fcntl(8, F_GETFD) = 0
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fcntl(10, F_GETFD) = 0
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fcntl(185, F_GETFD) = 0
fcntl(186, F_GETFD) = 0
fcntl(187, F_GETFD) = 0
fcntl(188, F_GETFD) = 0
fcntl(189, F_GETFD) = 0
fcntl(190, F_GETFD) = 0
fcntl(191, F_GETFD)
```

[illegible]

```
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x76d50f445cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
openat(2, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_AS, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "60d8c1fda000-60d8c20cc000 r--p 0"..., 1024) = 1024
read(3, "00 08:30 13628"..., 1024) = 1024
read(3, "00 00:30 13631"..., 1024) = 1024
read(3, "/usr/lib/x86_64-linux-gnu"..., 1024) = 1024
read(3, "0 13625"..., 1024) = 566
close(3) = 0
sched_getaffinity(0x3, 32, [0 1 2 3 4 5 6 7]) = 0
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x76d50f8f7000
mprotect(0x76d50f8f7000, 4096, PROT_READ) = 0
sigaltstack({ss_sp=0x76d50f8f8000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 963
rt_sigaction(SIGSEGV, {sa_handler=0x60d8c22ff0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x76d50f445cb0}, NULL, 0) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=0x60d8c22ff0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x76d50f445cb0}, NULL, 0) = 0
prctl(PR_SET_AUV, 0x7fff3c2b6c00, 0x200, 0, 0) = 440
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/ucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a"..., 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x76d50f445cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x76d50f445cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x76d50f445cb0}, {sa_handler=0x60d8c22ff0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x76d50f445cb0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x76d50f445cb0}, {sa_handler=0x60d8c22ff0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x76d50f445cb0}, 0) = 0
fcntl(4, F_GETFSZ, {c_lflag=LCNL|LXON, c_oflag=0x0|CB|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=0x0|CS0|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOL|ECHOK|EXTEN|ECHOCTL|ECHOKE, ...}) = 0
hbk(0x60d8c22b600) = 0x60d8c22b600
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_BASIC_STATS|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0777, stx_size=34, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0777, st_size=34, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0600, st_dev=makedev(0x0, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 34
splice(4, NULL, 1, NULL, 34, 0) = 34
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
```

OBSERVATION:

The strace utility was successfully used to trace the system calls generated by the cat sample.txt command. The trace showed system calls such as execve(), openat(), read(), write(), and close(). The execve() call was used to execute the cat program, while openat() opened the file for reading. The read() system call retrieved the contents of the file, and write() displayed the contents on the terminal. Finally, close() released the file resource. This demonstrates how a user-level command interacts with the Linux kernel through system calls to access files and perform I/O operations.